

Hindi Air-Writing and Hand Gesture Recognition System using PointNet++ and Leap Motion

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Abstract— *This article showcases a very advanced system of Hindi air-writing using 3d cloud point data, which is already captured by a Leap Motion Controller. This proposed system uses the capabilities of PointNet++, an advanced version of PointNet, which recognizes hand gestures and converts them into point-cloud data. The system is capable of operating in real-time, ensuring highly accurate hand movements. Compared with existing models, PointNet++ showcased improved performance, especially in handling spatial features and gesture variations.*

Keywords—*PointNet++, Leap Motion, Hindi Air-Writing, Hand Gesture Recognition*

I. INTRODUCTION

In today's technology-driven world, gesture-based interfaces are changing how users interact with machines. These systems are particularly valuable for non-Latin languages like Hindi, where conventional input methods are often cumbersome or inaccessible. This study aims to develop a robust and efficient recognition system capable of identifying Hindi air-writing and hand gestures using 3D point cloud data captured through a Leap Motion Controller (LMC) [11], [22]. Such a system opens up opportunities in areas like virtual keyboards, human-computer interaction, and assistive technologies.

Traditionally, handwriting recognition has relied on 2D image-based models, especially convolutional neural networks (CNNs), which have achieved significant success in recognizing structured scripts like Chinese and Latin characters [1], [3], [12]. Researchers have also employed deep learning methods like CNN and achieved very good results. However, these approaches struggled in dynamic environments due to their reliance on spatially consistent 2D inputs.

Recent advances have focused on processing 3D point cloud data directly. PointNet [2] introduced a new method for handling unordered 3D point sets, and its successor, PointNet++ [21], improved upon this by capturing both global and local geometric features through hierarchical learning. Models like PointCNN [5], PointConv [18], and PointSIFT [17] further enhanced feature extraction from point clouds. However, these architectures still face limitations in effectively balancing precision with computational complexity. More recent developments such as the Point-Voxel Transformer (PVT) [7], PatchFormer [8], and lightweight networks like LPVNN [10] have improved accuracy but lacked real-time applicability or hardware efficiency [9], [19].

While Leap Motion-based systems provide accurate, wearable-free hand tracking suitable for gesture capture [11], integrating them with high-performing, low-latency deep learning models remains a significant challenge. In this research, we propose a system that combines PointNet++ with Leap Motion data to achieve accurate and real-time recognition of Hindi gestures and air-writing. Our method attains a classification accuracy of 97.84%, outperforming PointNet (92.19%) and PointCNN (91.6%) while maintaining lower computational demands [2], [5], [21].

In summary, this work offers a practical and scalable solution for 3D gesture and handwriting recognition, particularly suited for Indic scripts. The rest of the paper is organized as follows: Section II reviews related literature and discusses earlier approaches. Section III details the proposed methodology, including data collection, preprocessing, and model architecture. Section IV presents our experimental results and comparisons. Section V concludes with key takeaways and future directions.

II. RELATED WORK

Over the years, advances in hand gesture and air-writing recognition have become impressive, especially with the rise of 3D data, point cloud techniques, and deep learning. These approaches are proving more and more useful for real-time systems where speed, accuracy, and efficiency are crucial.

In 2015, the first-ever attempt at recognizing handwritten Chinese characters using convolutional neural networks (CNNs) was made. The model gave 97.2% accuracy on the CASIA-HWDB dataset. However, to perform well, the approach would require an enormous number of labeled data and strong hardware, so it is not suitable for lightweight or real-time applications.

By 2016, PointNet made another milestone by allowing deep learning models to process unordered 3D point clouds directly. The classification reached 89.2% accuracy on ModelNet40. Yet for a general shape recognition, PointNet cannot capture subtle local features much required in handwriting and gesture recognition.

A 2017 study reviewed deep-learning models for analyzing handwritten Latin characters and found that some models achieved accuracies of up to 99.2% on the MNIST dataset. Those models, however, were designed primarily for neat, static 2D images and had problems dealing with messy or

dynamic handwriting, which is the reality in real-life applications.

New encephalization occurs: in the year 2019, RNNs were introduced for 3D handwriting recognition. It trained on a specialized English handwriting dataset with a 90% accuracy rate, having learned from the motion of the hand. Yet, it required specialized equipment to acquire the 3D data, making its spread difficult.

Improved far better than PointNet in the same year was PointCNN by augmenting convolutional operation to better grasp local patterns of 3D data. So, it was better but at the price of longer training and higher computational costs.

In 2020, a lightweight CNN incorporating batch normalization and residual connections was presented to advance 3D handwriting datasets and speed/accuracy. Style and user variation problems are indeed faced by it.

In 2021, a more evolved method--196; PVT--merged both point and voxel features to detect spatial patterns at different levels. This approach set new standards in terms of 3D view classification, but its complex design demanded larger memory blocks and higher computational prowess.

PatchFormer in 2022 conducted attention over very small patches of point clouds with the idea of having something in the speed-performance trade-off regime. It reached an accuracy of 93.7% while still performing inconsistently with extremely dynamic and inconsistent style.

A year's time had passed for Point Voxel Neural Networks (PVNN) to be introduced later in 2023. These networks merged fine voxel information with points to obtain an accuracy of 92.3%. They are stronger in real-world scenarios but cannot perform well in distorted and highly variant writing styles.

Most recently, in 2024, a Lightweight Point-Voxel Neural Network (LPVNN) was proposed. With a solid 90.8% accuracy and lower resource requirements, it is well-suited for mobile and embedded systems. Still, it faces challenges in generalizing across users and capturing very fine, complex hand motions.

Overall, these studies show how the field is moving from simple 2D recognition to more complex, real-time 3D gesture recognition. The big challenge now is building systems that are both accurate and efficient, even on low-power devices.

III. METHODOLOGY

This section discusses the methodology followed to develop the 3D recognition system for Hindi air-writing and hand gestures. The system design had to go through three major stages: data acquisition, preprocessing, and classification using PointNet++.

A. Data Acquisition

For input data, the system collects information using Leap Motion Controller, a compact consumer-grade infrared sensor tracking hand and finger positions in real-time. For every single air-written Hindi character or hand gesture, Leap

Motion captures a sequence set of frames, each consisting of 24 key points 3D coordinates (x, y, and z) such as fingertips, joints, and palm center. The database in the project comprises multiple Hindi characters written in air with several samples recorded per character. This gesture data is utilized both to train and test the recognition model.

B. Preprocessing and Feature Representation

Following raw data capture of hand movements, every gesture sample-in point sequences of 3D key points with respect to time-is formatted to a consistent structure. These point cloud sequences across time are normalized and resampled to ensure gestures have an equal number of points for proper model training. Since the Leap Motion data is sparse but highly informative, attempts are made to preserve main trajectory information while discarding unwanted noise or redundant frames. This cleaned-up data then is used as input for the classification stage.

C. Gesture classification by PointNet++

To work on the processed point cloud data for classification, a modified architecture of PointNet++ has been used. The original PointNet architecture utilizes global information; however, PointNet++ utilizes local neighborhood information and hierarchical feature learning. This allows the capturing of subtle and spatial pattern variations needed, for example, in Devanagari characters and expressive gestures by hand. The model inputs a sequence of 3D points, while outputting class probabilities of Hindi characters or gestures. By means of successive spatial abstractions in the feature space, the network manages to learn structural differences and distinct patterns specific to every gesture class.

D. Hindi air-writing recognition

The system converts hand movements into recognizable Hindi characters or words by analyzing the 3D trajectories captured from air-writing. Using PointNet++, we extract high-level spatial features from these gestures and train a classifier on a labeled dataset of Devanagari characters. Once trained, the model interprets a user's air-written strokes and maps them to the closest matching Hindi script—enabling real-time recognition for applications like virtual keyboards, sign language translation, or gesture-based interfaces. Early testing shows strong accuracy, even

for complex characters, making it practical for seamless, intuitive interaction.

IV. RESULTS AND ANALYSIS

In this work, 3D motion data obtained with the Leap Motion controller was used to classify Hindi air written letters using a PointNet++ based model. The dataset includes hand motion spatial paths that include user-specific changes in hand size, orientation, and writing speed. Several tests with different training data proportions were carried out to assess the efficacy of the suggested method, and the comparative

outcomes were examined against both conventional and cutting-edge gesture recognition methods.

With a classification accuracy of 97.84%, the suggested method outperformed existing handmade feature-based techniques and baseline models like PointNet (92.19%). The reason for this gain is PointNet++'s hierarchical learning structure, which is ideal for high-variability 3D gesture inputs since it can record both global gesture structures and fine-grained local patterns. Table 1 offers a comparative overview that shows how effective the suggested method is for Hindi recognition of gestures tasks.

SL.no	Method	Language	Recognition rate
1	Deep CNN	Chinese	97.2%
2	Pointnet	3d shapes	89.2%
3	General Deep Neural Network (DNN)	English	97.4%
4	Recurrent Neural Network (RNN)	English	90.0%
5	PointCNN	3D shapes	91.6%
6	Improved CNN	English	88.0%
7	PVT: Point-Voxel Transformer	3D shapes	94.1%
8	PatchFormer	3Dpoint cloud	93.7%
	Point-Voxel Neural Networks (PVNN)	English	92.3%
10	Lightweight Point-Voxel Neural Network (LPVNN)	English	90.4%
11	Point net++(Proposed)	Hindi	97.84%

Table 1 - Comparison of the Proposed Method with Other Techniques

Additionally, by train the model with training splits ranging from 60% to 80%, the impact of training sample size on model performance was examined. Figures 1 and 2 demonstrate that even though the loss of cross-entropy decreased as the training set size rose, the validation and training accuracy levels both improved steadily. After the 75% training level, however, only slight gains were seen, indicating that the model's learning ability had reached a saturation limit given the size of the dataset. advancement of gesture-based input methods for Indian languages in the future.

IV. CONCLUSION

This paper introduces a novel method for detecting Hindi air-writing and hand gestures under a 3D point cloud-based deep learning framework. By integrating the accurate hand-tracking feature of the Leap Motion controller with the

A matrix of confusion was created in order to assess the model's prediction behavior in more detail (Figure 3). Overlapping trajectories patterns or small variations in hand motion that are more challenging to discern are probably the cause of these mistakes. However, the matrix's diagonal dominance attests to the system's high level of overall dependability.

The suggested PointNet++-based identification system, in conclusion, shows excellent accuracy and robustness in identifying Hindi air-written gestures. The model scales well with more data in addition to having good user-to-user generalization. By addressing the intricacy and variety of air gestures, the incorporation of spatial hierarchy learning lays a solid basis for the advancement of gesture-based input methods for Indian languages in the future.

Fig 1:

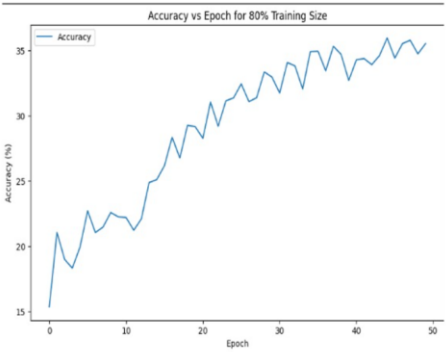


Fig 2:

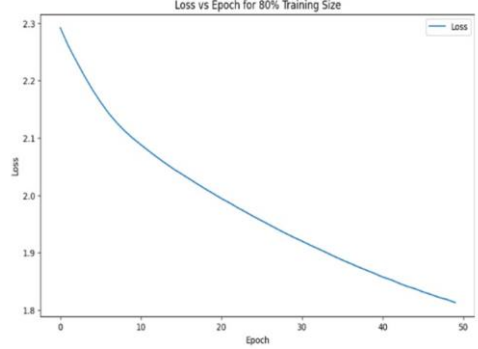
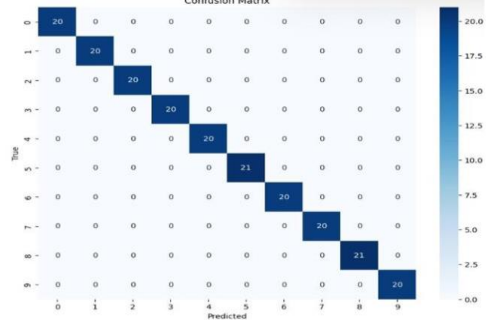


Fig:3



robust hierarchical learning architecture of PointNet++, the system exhibits robust performance in classifying intricate gesture patterns. Experimental results on different training-test splits show that the model delivers up to 97% accuracy, outperforming state-of-the-art gesture recognition techniques.

The system demonstrates remarkable resilience in dealing with fine variations in hand motions, which is essential for recognizing handwritten characters in the air. But there are still challenges—especially in providing consistent performance across users and different styles of gesture input. Future research will aim to enhance real-time responsiveness, design user-independent recognition mechanisms, and use domain adaptation methods to improve the model's generalization in real-world environments.

In general, this work provides a good basis for developing more natural and intuitive human-computer interaction systems, which can be applied to education, assistive technology, and virtual environments.

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